Math 1200 Lecture 5

Sep 17 2026

Topics: 14.1-14.3: Functions of several variables, partial derivatives Name:_____
Collaborators:_____

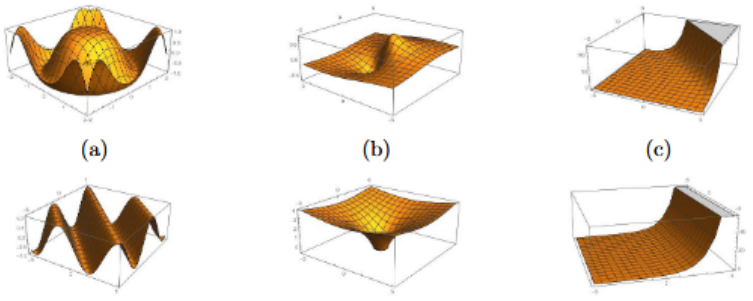
Group work 1 Find and sketch the domains of the functions. As a bonus, also sketch their contour plots.

(a) $f(x, y) = \sqrt{x + y}$

(b) $f(x, y) = \ln(xy)$

Group work 2 Match the functions with their graphs and contour plots.

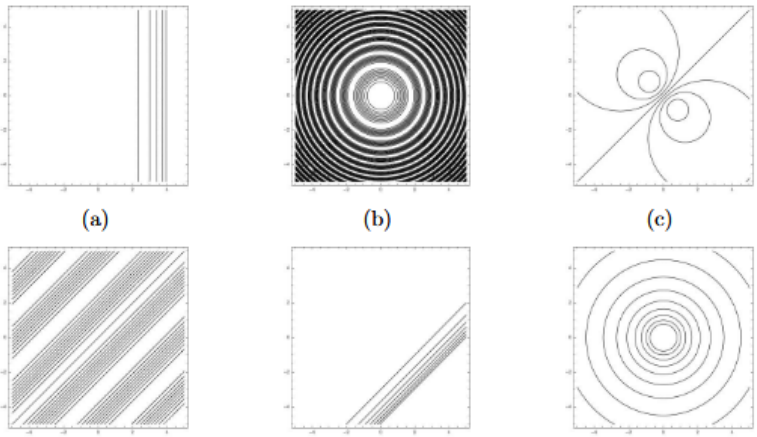
1. $f(x,y) = e^{x-y}$: Graph _, Contour _
2. $f(x,y) = e^{x}$: Graph _, Contour _
3. $f(x,y) = \cos(x-y)$: Graph _, Contour _
4. $f(x,y) = \ln(x^2+y^2)$: Graph _, Contour _.
5. $f(x,y) = \frac{x-y}{1+x^2+y^2}$: Graph _, Contour _.
6. $f(x,y) = \cos(x^2+y^2)$: Graph _, Contour _



(a) (b) (c)

(d) (e) (f)

Figure 1: The graphs for problem



(a) (b) (c)

(d) (e) (f)

Figure 2: The contours for problem

Group work 3 Compute the second partial derivatives of the function $f(x, y) = e^{xy}$.

a) $f_{xx} =$

b) $f_{xy} =$

c) $f_{yy} =$

d) $f_{yx} =$

Group work 4 Calculate $\frac{\partial}{\partial x}\frac{\partial}{\partial x}\frac{\partial}{\partial y}f(x, y)$ for

$$f(x, y) = e^{y\sin(y^{10})} + x\frac{y(y+4)(y+10)}{\ln(\sin(\cos(y+y^2)))} + x^2y$$